International Communications in Heat and Mass Transfer 172 (2026) 110139

Contents lists available at ScienceDirect

International Communications in Heat and Mass Transfer

journal homepage: www.elsevier.com/locate/ichmt

Repeatability of heat transfer coefficient and critical heat flux for flow boiling of water in copper microchannels part 1: Experimental investigation

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A R T I C L E I N F O

Keywords:
Flow boiling
Copper microchannel
Ageing
Critical heat flux
Heat transfer coefficient

A B S T R A C T

Two-phase cooling with microchannels offers very high heat transfer coefficients and also help to maintain fairly constant device temperature, and hence a promising solution for electronics cooling. This work reports the experimental investigation of the repeatability of flow boiling heat transfer coefficient and critical heat flux for two copper channels – one is 1.435 mm wide, 0.712 mm deep and 40 mm long with a hydraulic diameter of 0.952 mm and the other is 1.984 mm wide, 0.85 mm deep and 40 mm long with a hydraulic diameter of 1.19 mm. Maximum reductions in two-phase heat transfer coefficient of 52% and 53% were observed for the 0.952 mm and 1.19 mm channels, respectively. At the same time, the CHF was found to be enhanced by up to 85% and 78% for the 0.952 mm and 1.19 mm channels, respectively due to ageing. The EDS examination of the channel copper surface demonstrates thermal oxidation during ageing, while the SEM and confocal microscopy investigations illustrate the change in surface morphology caused by ageing. Also, the in-situ contact angle measurements indicate that the channel surface becomes hydrophilic due to ageing. The reasons for the variations in heat transfer coefficient and CHF are discussed.

1. Introduction

Thermal management of devices operating under high heat flux conditions such as microprocessors, insulated gate bipolar transistors (IGBTs), etc. is challenging due to the limited space available. Two-phase and single-phase liquid cooling employing microchannels are promising solutions for such applications. Two-phase cooling enables significantly higher heat transfer coefficient (HTC) in comparison with single-phase cooling for the same mass flow rate of coolant. Critical heat flux (CHF) is one parameter that restricts the heat rejection capacity of a two-phase cooling system. During flow boiling in microchannels, CHF can occur as a result of departure from nucleate boiling (DNB), film dryout, or intermittent dryout.

Surface characteristics such as wettability and surface microstructure can affect the two-phase HTC and CHF. Generally, a hydrophilic surface promotes wetting and enhances CHF while the superheat necessary for nucleation is reduced by a hydrophobic surface. An experimental investigation was performed by Karim et al. [1] on flow boiling in a microgap heat sink, which had a hydrophilic heated surface obtained by sputtering SiO_2. They observed that the hydrophilic surface was able to mitigate intermittent dryout observed in the case of a microgap heat sink without SiO_2 coating. Higher average heat transfer coefficients as well as lower hot spot temperature were also reported for the hydrophilic channel as compared to uncoated channel. Ahmadi et al. [2] investigated the influence of channel surfaces having a wettability gradient during flow boiling in a microchannel. The channels were prepared by depositing Al_2O_3 over SiO_2 formed on a silicon wafer using photolithography. They observed that the surfaces with wettability gradient could promote slug flow regime over bubbly flow. This led to HTCs in comparison to a channel with plain hydrophobic surface. An experimental investigation was performed by Aboubakri et al. [3] on the influence of surfaces with mixed-wettability during flow boiling in a microchannel. They employed FC-72 as the working fluid. The microchannel surface had three regions from inlet to outlet. All three regions had hydrophobic islands over a hydrophilic base. The area covered by hydrophobic islands was highest in the region near the inlet and then decreased in the two regions downstream. They reported a maximum increase in HTC of 50%. The enhancement was explained by a higher number of nucleation sites and break-up of the vapour bubbles due to the hydrophobic islands. An experimental investigation was performed by Zhou et al. [4] on the influence of the wettability of microchannel surface during flow boiling. Three different surfaces having contact angles of 70° (untreated), 43° (hydrophilic) and 0° (superhydrophilic)

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https://doi.org/10.1016/j.icheatmasstransfer.2025.110139

Available online 2 December 2025

Nomenclature

a	channel width, m
A	area of cross-section, m^2
b	channel height, m
C_p	specific heat J/kg K
Co	Confinement number, $Co = (\sigma/g\Delta\rho)^{0.5}/d_h$
d_e	equivalent diameter, $d_e = 4b$, m
d_h	hydraulic diameter, $d_h = 2ab/(a+b)$, m
d_{th}	threshold diameter, $(\sigma/g\Delta\rho)^{0.5}/Co$, m
f	Fanning friction factor
g	acceleration due to gravity, m/s^2
G	mass flux, kg/m^2s
h	heat transfer coefficient, W/m^2K
h_{fg}	latent heat of vaporization, J/kg
I	heater current, A
k	thermal conductivity, W/mK
L	channel length, m
$\dot{m}$	mass flow rate, kg/s
P	pressure, Pa
P_h	heated perimeter, Ph = a, m
ΔP	pressure drop, Pa
q	heat flux, W/m^2
Q	heat rate, W
R_a	arithmetic mean roughness, μm
R_q	root mean square roughness, μm
R_p	maximum peak height, μm
R_v	maximum valley depth, μm
R_t	Maximum profile height, $R_t = R_p + R_v$
Re	Reynolds number, $Re = G\, d_h/\mu$
T	Temperature, K
V	heater voltage, V
We	Weber number, $We = G^2L/\sigma\, \rho_f$
x	thermodynamic equilibrium quality
z	distance, m

Greek symbols

σ	surface tension, N/m
α	channel aspect ratio, $\alpha = b/a$
μ	viscosity, kg/ms
ρ	density, kg/m^3

Subscripts

amb	ambient
app	apparent
ch	channel
CHF	critical heat flux
exp	experimental
fd	fully developed
g	vapour
h	hydraulic
i	channel inlet
l	liquid
o	channel outlet
pred	predicted
p, i	plenum inlet
sat	saturated
sub	sub-cooled
th	threshold
tp	two-phase
tw	thermocouple beneath bottom of channel
w	wall

Abbreviations

CHF	critical heat flux
HTC	Heat transfer coefficient
MAE	mean absolute error

were investigated. They observed that the superhydrophilic surface had maximum number of nucleation sites due to a higher surface roughness. The departure frequency of bubbles was also higher for the superhydrophilic surface. The average HTC on superhydrophilic surface was enhanced up to 325% in comparison with a surface which was bare. Dryout was avoided in the case of the superhydrophilic surface unlike the untreated surface. This was explained by the higher surface tension force acting on the liquid film in the case of superhydrophilic surface. Wang et al. [5] studied the effect of pitch of hydrophobic spots in wettability patterned microchannel during flow boiling. They observed that larger pitch for hydrophobic spots gave higher HTC at lower mass fluxes due to bubble coalescence during nucleation in the case of smaller pitch. They reported that a higher pitch also led to higher CHF due to better rewetting. A lower pitch increased the flow resistance due to presence of larger bubbles and more number of nucleation sites. An experimental investigation was performed by Nedaei et al. [6] on the influence of wettability gradient surfaces during flow boiling in microtubes. They used tubes with wettability varying from hydrophobic to hydrophilic on moving from inlet to outlet and vice versa. A maximum increase in HTC of 64% was observed for the case with hydrophobic inlet and hydrophilic outlet and the enhancement was explained by increased nucleation site density and bubble departure.

Mathew et al. [7] studied the influence of periodic stepped fins whose height was reduced to half the total fin height. The stepped fins facilitated growth of bubbles in transverse direction and minimized flow reversal. They also provided a higher nucleation site density. Consequently, an enhanced heat transfer performance was obtained as opposed to straight finned microchannel at low to moderate heat fluxes. CHF was also enhanced at low mass fluxes. Chang et al. [8] investigated the combined influence of micro-pin fins along the sidewalls and silicon nanowires grown on all inside surfaces of the microchannel on flow boiling heat transfer. The design promoted capillary flow and led to very high frequency of rewetting. They reported a maximum enhancement in CHF and HTC of 483% and 122%, respectively. Woodcock et al. [9] investigated flow boiling in a minichannel with bi-layered micro-pin fins employing HFE7000 as the working fluid. They observed that heat fluxes up to 1 kW/cm^2 could be dissipated maintaining the device temperature below 95^o C. Deng et al. [10] employed open-ring pin fins in a microchannel. They reported that wall superheat required for ONB was reduced to 1^o-2^o C due to nucleation sites provided by the pin fin structures. An experimental investigation was performed by Deng et al. [11] on the influence of transverse interconnects during flow boiling of water in parallel re-entrant microchannels. They observed that the wall superheat required for ONB was reduced to 0.4 °C– 1.2 °C due to nucleation sites present in interconnecting channels. The HTC was enhanced by up to 280% in comparison with a parallel re-entrant microchannel without interconnects. The enhancement was attributed to the alternate pathway provided by the interconnects for vapour bubble growth which also helped to reduce the bubble length. Khanikar et al. [12] studied the influence of CNTs on the microchannel bottom wall on two-phase HTC and CHF. They observed that the CHF was enhanced initially because of the higher heat transfer area offered by the CNTs but after repeated experiments at higher mass fluxes, the CNTs bent which led to a reduction in the enhancement. They also reported that the HTC increased after deformation of CNTs due to generation of more nucleation sites. An experimental investigation was performed by Yin et al. [13] on the influence of a hybrid microstructured surface during flow boiling in a microgap heat sink. The inlet side of the microstructured surface was roughened. At the same time, a micro pin fin array was provided on the outlet side. An improved heat transfer performance was obtained as a consequence of the suppression of local dryout by the micro pin fins and increased the nucleation site density in the roughened region. An experimental investigation was performed by Luo et al. [14] on the influence of microporous decorated sidewalls on flow boiling in silicon microchannels. They observed a maximum enhancement in CHF of 205% and a maximum enhancement in HTC of 228%. The increase in CHF and HTC was attributed to delay in local dryout caused by extended retention of liquid film and thin film evaporation, respectively. An experimental investigation was performed by Lee et al. [15] on the influence of a porous coating on flow boiling of HFE-7200 in a copper microchannel. An increase in two-phase HTC up to 44% was observed for the coated channel due to increased density of nucleation sites. The increase in nucleation site density was a consequence of deep cavities present on the coated surface. Jones and Garimella [16] investigated flow boiling in parallel microchannels having roughness values of 6.7 μm, 3.9 μm, and 1.4 μm. The channels were created employing EDM and saw cutting. They observed that HTCs were higher for channels with roughness values of 6.7 μm and 3.9 μm at heat fluxes exceeding 1500 kW/m^2. The differences in HTC were very less at heat fluxes below 700 kW/m^2. Additionally, roughness had little effect on CHF.

An experimental investigation was performed by Bhide et al. [17] on flow boiling of water in microchannels of diameters 0.065, 0.07, 0.107 and 0.125 mm. They observed that the heat flux at ONB increases with diameter for the same mass flux. Hong et al. [18] investigated flow boiling of ethanol and acetone in shallow rectangular microchannels of diameters 0.5 mm and 0.667 mm. They observed that the CHF reduced with an increase in diameter. Basu et al. [19] investigated flow boiling in circular microtubes of diameters 1.6 mm, 0.96 mm and 0.5 mm. The working fluid was R134a. They observed an increase in CHF with an increase in tube diameter. Ong and Thome [20] investigated flow boiling in narrow tubes of diameters 3.04 mm, 2.2 mm, 1.03 mm, 0.79 mm, and 0.51 mm. R245fa, R236fa and R134a were used as the working fluids. They observed that the CHF was maximum for the tubes of diameters 0.79 mm and 1.03 mm and decreased for diameters greater than 1.03 mm and also for diameters lower than 0.79 mm.

Materials possessing high thermal diffusivity, high thermal conductivity, and good machinability are required for heat sinks. All of these can be seen in copper. During flow boiling of water in copper microchannels, ageing or thermal oxidation of copper occurs [21–23]. Ramakrishnan et al. [24] reported that the oxide layer formed increases the wettability of the channel and leads to reduction in HTC [24]. They also found that, for high heat flux boiling in a copper microchannel of hydraulic diameter 0.641 mm, CHF was not significantly affected due to ageing. The objective of the present experimental study is to examine whether ageing affects CHF during flow boiling in channels of different diameters and also to examine the extent of change in HTC.

2. Experimental procedure and details

The experimental test rig used in the present investigation is the same as that by Ramakrishnan et al. [24]. Fig. 1 shows a schematic of the test rig. DI water is used as the working fluid. A coarse 100 μm inline filter placed after the reservoir traps dust particles. A membrane-contractor type degasifier connected to a vacuum pump removes the dissolved gases from the working fluid. The working fluid is driven

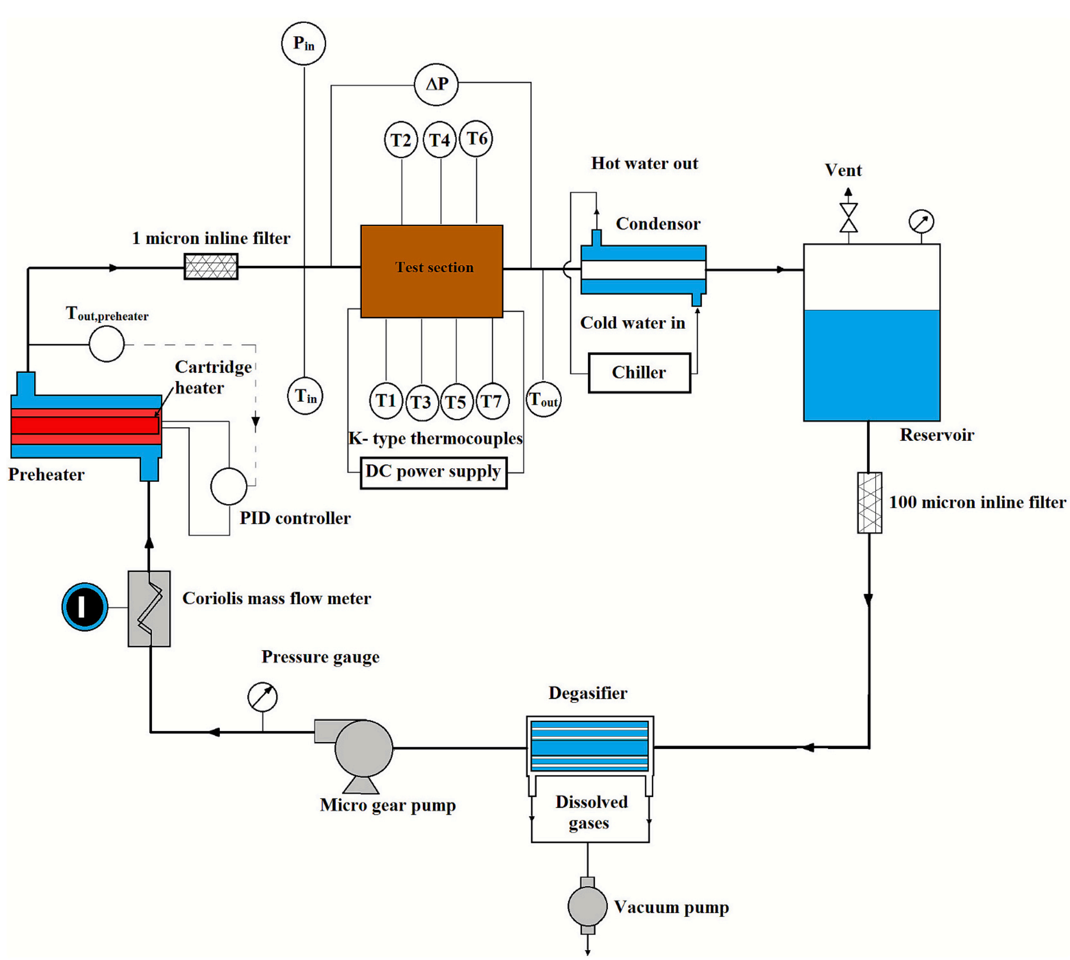


Fig. 1. Experimental circuit.

through the loop employing a micro-gear pump. A Coriolis mass flow meter measures the flow rate. A preheater controlled by a PID controller maintains the required inlet temperature of water in the test section. To ensure that fine particles do not accumulate and block the microchannel, a 0.5 μm inline filter is used. The test section is connected to a differential pressure transmitter, pressure sensors and thermocouples. The working fluid in two-phase state passes through a condenser before entering the reservoir. Ageing experiments were performed on the two freshly machined test sections. In the case of the test section having hydraulic diameter of 0.952 mm, the ageing experiments were performed at the mass flux of 500 kg/m^{2}s. In the case of the test section of hydraulic diameter 1.19 mm, the ageing experiments were performed at the mass flux of 300 kg/m^{2}s. The experiments were started with the mass fluxes maintained at these values. The DI water inlet temperature was set at 92 °C. The cartridge heater inside the test section was supplied with power that was increased stepwise. Following each power increment, the system attained steady state conditions in a short while and then data was saved for 10 min. The power input to the test section was raised till the temperature indicated by thermocouples mounted in wall reached 160 °C. Subsequently the power supply was shut down. The ageing experiments were repeated until the relative difference between the HTCs obtained from three consecutive experimental runs was less than 10%. On completion of the ageing experiments, two-phase experiments were conducted to investigate the influence of mass flux on CHF on the fully aged channel. The mass fluxes used for the three test sections are presented in Table 1. These mass fluxes are the values maintained at the beginning of the experiment. During these experiments also, power was given to the test section incrementally. The dissolved oxygen concentration in water in the experimental circuit was estimated employing a HQ30D portable oxygen meter. The dissolved oxygen concentration measured was within 2 PPM.

2.1. Test section details

Two rectangular microchannels of different dimensions are investigated in the present study. The microchannel dimensions are presented in Table 1. Each test section comprises a heated copper block, a PEEK cover plate with microchannel and a teflon housing. All components were fabricated by CNC milling. The microchannel has been machined on PEEK material because of its capacity to withstand elevated temperatures up to 240 °C without deformation. At the same time, since PEEK is opaque in nature, flow visualization cannot be done. The microchannel has been machined on PEEK and not copper because it allows in-situ contact angle measurement on copper. The only heated surface of the channel is formed by the top flat surface of the copper block. The copper block was made employing CNC end milling. Inserts of diameter 5 mm and made out of teflon are provided in the copper block to prevent heat transfer to the working fluid at the outlet plenum and inlet plenum. The PEEK cover plate is fixed strongly with the copper block using SS Allen screws. The copper block was provided with threaded holes for inserting the screws. Thermocouples having diameter of 1 mm and of K-type are used for temperature measurement. A total of seven thermocouples are fixed lengthwise of the microchannel at a depth of 3.5 mm from the bottom of the microchannel. An electric cartridge heater inserted into the copper block is used to heat the test section. Thermal paste is coated around the heater to reduce thermal resistance caused by air gaps. The test section design features are shown in Fig. 2. A photograph of the assembled test section is shown in Fig. 3.

2.2. Data reduction

The data reduction method followed in the present study is similar to the method followed by Ramakrishnan et al. [24]. First, single-phase experiments are performed to calculate heat loss to the ambient from the test section. The difference in enthalpy of the fluid is estimated as it passes through the microchannel.

$$Q_{loss} = Q_{total} - \dot{m}C_p(T_o - T_i) \tag{1}$$

$$Q_{total} = VI \tag{2}$$

The heat loss can subsequently be represented as function of difference of mean thermocouple measured wall temperature and ambient temperature $(\overline{T_{tw}} - T_{amb})$. A graph of heat loss against $(\overline{T_{tw}} - T_{amb})$ is made as presented in Fig. 4. During flow boiling experiments where the wall temperatures are higher, heat loss can be estimated by extrapolating the linear fit between heat loss and $(\overline{T_{tw}} - T_{amb})$. Similar method has been reported in literature [25–27].

By deducting the heat loss from total power supplied, effective power input to fluid can be estimated.

$$Q_{fluid} = Q_{total} - Q_{loss} \tag{3}$$

Wall heat flux is given by the ratio of the effective power input to the channel heated area.

$$q = \frac{Q_{fluid}}{A_S} \tag{4}$$

$$A_S = aL \tag{5}$$

The thermocouple measured mean wall temperature is given by the mean of the readings from the seven thermocouples fixed 3.5 mm below the channel heated surface.

$$\overline{T_{tw}} = \frac{1}{7}\sum_{i=1}^{7} T_{tw,i} \tag{6}$$

Fourier's law is employed to evaluate the local wall temperatures in all seven locations lengthwise of the channel, assuming 1-D conduction [16,28,29].

$$T_{w,i} = T_{tw,i} - q\frac{\Delta y}{k} \tag{7}$$

k denotes the thermal conductivity of copper and the value is taken as 385 W/mK, and $\Delta y = 0.0035$ m.

The inlet temperature of liquid during flow boiling experiments is set at 92 °C. At the channel inlet, the liquid is subcooled and at some position along the microchannel, the thermodynamic quality (x) of water attains the value zero. This position is the point of saturation. The two-phase HTC is calculated in the region of the microchannel wherein the fluid is saturated. By subtracting the subcooled length from the channel length, the saturated length can be calculated.

$$L_{sat} = L_{ch} - L_{sub} \tag{8}$$

By iteratively solving the equations for pressure drop and energy balance, the subcooled length can be evaluated. The subcooled length is calculated using energy balance as given below:

$$L_{sub} = \frac{\dot{m}c_p\left(T_{sat}(z_{sub}) - T_i\right)}{qa} \tag{9}$$

Table 1
Dimensions of the channels investigated and mass fluxes used in experiments with aged channel.

Test section no.	Length of channel (mm)	Width of channel (mm)	Depth of channel (mm)	Hydraulic diameter (mm)	Standard deviation in measured dimensions (%)	Aspect ratio	Mass fluxes (nominal) (kg/m^{2}s)
1	40	1.435	0.712	0.952	0.4	0.496	300, 400
2	40	1.984	0.85	1.19	0.5	0.428	400, 200

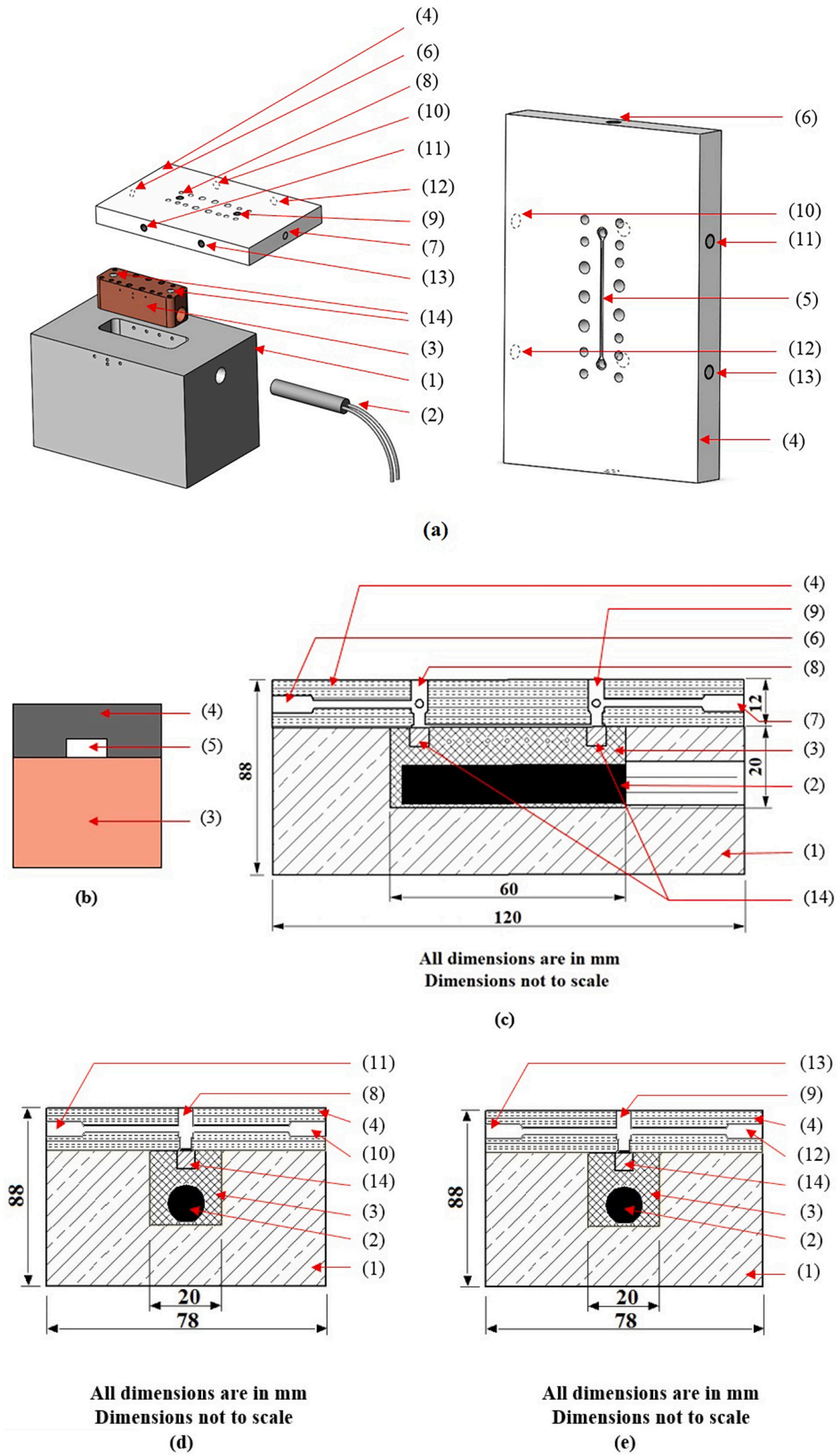


Fig. 2. Test section features.
(a) Exploded view of test section, (b) Widthwise cut section view of PEEK microchannel (Not to scale), (c) Longitudinal cut section view of test section assembly, (d) Widthwise cut section view of test section assembly at microchannel inlet, (e) Widthwise cut section view of test section assembly at microchannel outlet. **Legend**: (1) Teflon housing, (2) Cartridge heater, (3) Heated copper block, (4) PEEK cover plate, (5) Microchannel, (6) Liquid inlet, (7) Liquid outlet, (8) Inlet pressure sensor tapping (9) Outlet pressure sensor tapping, (10) Tapping for inlet leg of differential pressure transmitter, (11) Hole for inserting inlet thermocouple, (12) Tapping for outlet leg of differential pressure transmitter, (13) Hole for inserting outlet thermocouple, (14) Teflon inserts.

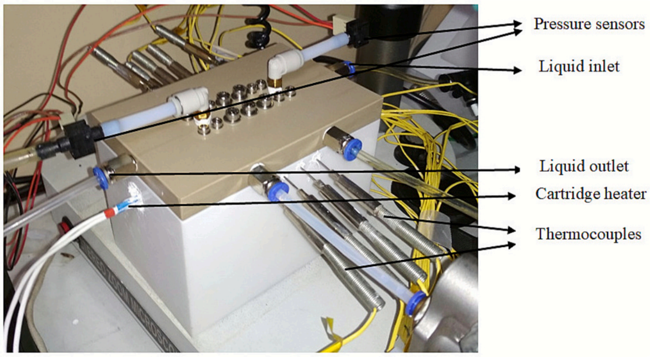


Fig. 3. Assembled test section connected in experimental circuit.

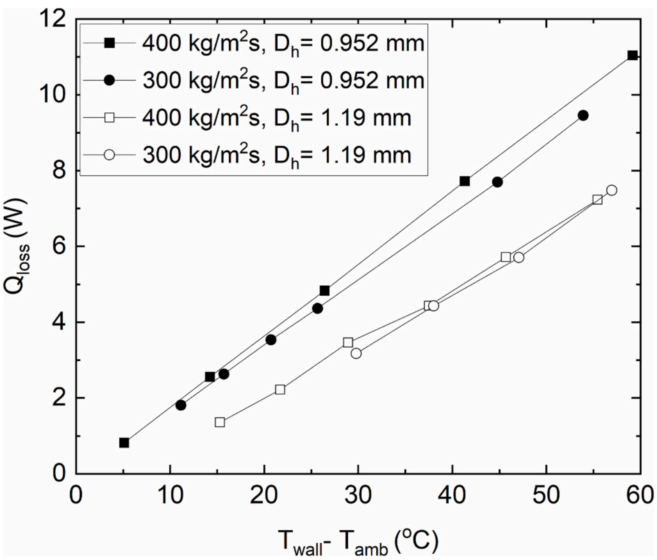


Fig. 4. Variation of heat loss with the difference between thermocouple measured mean wall temperature and the surrounding temperature.

The pressure drop across the subcooled length has to be evaluated and then deducted from the channel inlet pressure for evaluating the pressure at the point of saturation.

$$p_{sat}(z_{sub}) = p_i - \frac{2 f_{app} G^2}{\rho_l D_h} L_{sub} \tag{10}$$

The Shah [30] correlation is employed to calculate the apparent friction factor $\left(f_{app}\right)$. First, the value of L_{sub} is evaluated from Eq. (9) by using an initial guess value for $T_{sat}(z_{sub})$. Then, $p_{sat}(z_{sub})$ is evaluated by substituting this value of L_{sub} in Eq. (10). Then, the saturation temperature $T_{sat}(z_{sub})$ corresponding to $p_{sat}(z_{sub})$ is derived from Bell et al. [31]. The Eqs. (9) and (10) are solved iteratively till the values of $T_{sat}(z_{sub})$ derived from both equations are within 0.01% of each other. There are studies in literature [16,27,28] which use this procedure for evaluating subcooled length.

The pressure drop between the inlet and outlet plenums is used to measure the pressure drop across the microchannel. While estimating the total pressure drop across the channel, the losses due to contraction and expansion at the inlet and outlet plenums need to be estimated. The losses are estimated following the equations proposed by Lee and Garimella [28].

$$\Delta P_{ch} = \Delta P_{measured} - \left(\Delta P_{cont} - \Delta P_{exp}\right) \tag{11}$$

The difference between pressure indicated by inlet pressure sensor and pressure loss due to contraction gives the pressure at the channel inlet.

$$P_i = P_{p,i} - \Delta P_{cont} \tag{12}$$

The difference of inlet pressure and channel pressure drop gives the pressure at channel outlet.

$$P_o = P_i - \Delta P_{ch} \tag{13}$$

In the saturated region of the microchannel, local pressure is assumed to decrease in a linear manner from the point of saturation till the exit of the microchannel. The following equation is employed in the saturated region for calculating the local pressure [27].

$$P_{sat}(z) = P_O + \frac{L - z}{L - L_{sub}} \left(P_{sat}(z_{sub}) - P_o\right) \tag{14}$$

The two-phase pressure drop (ΔP_{tp}) is the pressure drop along the saturated region of the microchannel. It is obtained by reducing the channel outlet pressure from the pressure at the point of saturation.

$$\Delta P_{tp} = P_{sat}(z_{sub}) - P_o \tag{15}$$

The mean wall superheat ($\overline{T_w - T_{sat}}$) in the saturated region of the microchannel has to be evaluated for estimating the two-phase HTC.

$$\overline{T_w - T_{sat}} = \frac{1}{n} \sum_{i=1}^{i=n} \left(T_{w,i} - T_{sat,i}\right) \tag{16}$$

In the above equation, the local saturation temperature ($T_{sat,i}$) is evaluated using Eq. (14) and the local wall temperature ($T_{w,i}$) is evaluated using Eq. (7).

The mean two-phase heat transfer coefficient is determined as follows.

$$h_{tp} = \frac{q}{\overline{T_w - T_{sat}}} \tag{17}$$

Based on energy balance, the thermodynamic exit quality is calculated as follows.

$$x_o = \frac{1}{h_{fg}} \left[\frac{Q_{fluid}}{\dot{m}} - c_p\left(T_{sat}(z_{sub}) - T_i\right)\right] \tag{18}$$

The mean of outlet temperature and temperature at the point of saturation is used to calculate the enthalpy of vaporization of water in the above equation.

2.3. Uncertainty analysis

The present work reports quantities which comprise of measured quantities and derived quantities. The derived quantities can be expressed as a function of the measured quantities. Measured quantities have random error and systematic error, which need to be accounted for. The method suggested by Moffat [32] is used calculate uncertainty in derived quantities. Consider a derived quantity R which can be represented as function of a number of measured quantities x_i.

$$R = R\left(x_1, x_2, x_3, x_4 \ldots\ldots x_n\right) \tag{19}$$

For each measured quantity x_i, let U_{xi} denote the corresponding uncertainty. The following equation is employed to estimate the uncertainty in R.

$$U_R = \pm\sqrt{\sum_{i=1}^{n}\left(\frac{\partial R}{\partial x_i} U_{xi}\right)^2} \tag{20}$$

Uncertainties of derived quantities including two-phase HTC, pressure drop, single-phase friction factor, heat flux and mass flux have been calculated employing the above technique. The estimated values are shown in Table 2.

Table 2
Uncertainties of the measured and derived quantities.

Quantity	Uncertainty	
	TS #1	TS #2
Microchannel width	±0.007 mm	±0.007 mm
Microchannel depth	±0.005 mm	±0.005 mm
Surface profilometer	±0.03 nm	±0.03 nm
Power input, voltage	±0.5 V	±0.5 V
Power input, current	±0.005 A	±0.005 A
Temperature	±0.6 K	±0.6 K
Mass flow rate	± 0.1 g/min	± 0.1 g/min
Heat flux	±0.7% – 2.3%	±0.65%–1.67%
Mass flux	±0.91%–1%	±0.9%–0.94%
Single-phase friction factor	±4.6% - 13.3%	±6.4%–26.4%
Two-phase heat transfer coefficient	±4%–15.5%	±3.7%–12%
Two-phase pressure drop	±2% - 21%	±2%–56%

2.4. Surface characterization

Surface characterization was done on the channel prior to and after ageing experiments. They include dimension measurement of PEEK microchannel, Scanning Electron Microscope (SEM) imaging, elemental analysis using Energy Dispersive X-ray Spectroscopy (EDS), roughness measurement, confocal microscopy, and contact angle measurement (in-situ) over the copper block which forms the heated side of the channel. The depth and width of the channel were estimated making use of a Bruker Contour GT-1 optical profilometer at four sites lengthwise of the channel. The nominal dimensions of the channels are shown in Table 1. The standard deviations of measured dimensions are given in Table 1. Roughness measurements were also carried out employing a Bruker Contour GT-1 optical profilometer. An Olympus LEXT OLS 4000 microscope was employed for confocal microscopy. A Holmarc goniometer based on Sessile droplet method was used to determine the contact angle on the channel copper surface. The droplet volume was chosen as 1 μl. An FEI Inspect F scanning electron microscope was used to capture the channel copper surface images at high magnification. Surface chemistry analysis (EDS analysis) was done employing EDAX detector connected to the SEM. The oxide layer thickness on the aged test section was estimated using Bruker Contour GT-1 optical profilometer.

Roughness measurements were performed at different sites on the channel copper surface before and after ageing experiments on test sections 1 and 2 (0.952 mm channel and 1.19 mm channel) and the results are shown in Table 3. The average roughness is found to increase on 0.952 mm channel from 0.259 μm to 0.656 μm and on 1.19 mm channel from 0.168 μm to 1.323 μm. The increase in roughness can be attributed to the change in morphology of the copper surface due to ageing. 3-D roughness profiles captured after measurement of roughness are shown in Fig. E1 (presented in e-component). Copper surface images obtained using confocal microscope prior to and after ageing for the two test sections are presented in Fig. 5. The figure reveals the variation in surface texture due to ageing. The channel copper surface images taken using SEM before and after ageing experiments are shown in Fig. 6. It can be seen that the surface morphology has been altered due to ageing. EDS analysis was conducted before and after ageing to obtain the elemental composition of the heated copper surface. The results are shown in Fig. 7 and they point to the copper surface oxidation consequent to ageing. The oxygen content (atomic percentage) on the first test section increased from 6% before ageing to 37.6% after ageing while oxygen content (atomic percentage) on the second test section increased from 6.7% to 37.9%. In-situ measurement of contact angle was performed over the copper surface prior to and after the ageing experiments to analyze the influence of ageing on wettability of the copper surface. For measuring contact angle, the copper block was oriented such that the baseline of the droplet viewed by the camera was along the channel length. The results are shown in Figs. 8 and 9. The results show that the contact angle decreases or wettability increases as a result of ageing. The mean contact angle at five points lengthwise of the channel decreased from 96° to 22° for the first channel. For the second channel, the mean value of contact angle decreased from 97° to 29°. The arrangement of the copper block for measuring contact angle is shown in Figs. E2 and E3 (presented in e-component). The mean oxide layer thickness as estimated from the measurements performed at three axial locations is 6.83 μm for the 0.952 mm channel and 7.06 μm for the 1.19 mm channel. The oxide layer appears as a step or projection above the copper surface. The average height of the step above the copper surface gives the oxide layer thickness. A sample measurement of oxide layer thickness at a particular location of the channel is given in Fig. E4 (presented in e-component).

Table 3
Roughness measurements.

Case	Location	R_a (μm)	R_q (μm)	R_t (μm)
First channel (before ageing experiments)	1	0.224	0.302	5.2
	2	0.242	0.316	4.96
	3	0.247	0.316	4.75
	4	0.268	0.337	4.22
	5	0.288	0.362	4.96
	6	0.286	0.357	5.4
	Average	0.259	0.331	4.915
	Standard deviation	±0.0257	±0.0243	±0.407
First channel (after ageing experiments)	1	0.447	0.562	7.13
	2	0.608	0.833	10.71
	3	0.643	0.91	10.72
	4	0.716	0.982	10.27
	5	0.804	1.1	11.29
	6	0.721	1.3	19.99
	Average	0.656	0.947	11.685
	Standard deviation	0.123	0.249	4.33
Second channel (before ageing experiments)	1	0.198	0.356	13.93
	2	0.18	0.366	9.35
	3	0.176	0.217	4.5
	4	0.156	0.205	6.91
	5	0.155	0.195	6.55
	6	0.147	0.189	5.18
	Average	0.168	0.255	7.737
	Standard deviation	0.019	0.082	3.465
Second channel (after ageing experiments)	1	1.03	1.56	18.49
	2	1.28	1.79	23.03
	3	1.41	1.96	21.97
	4	1.45	1.83	15.36
	5	1.45	1.83	15.38
	6	1.32	1.71	18.51
	Average	1.323	1.78	18.79
	Standard deviation	0.159	0.135	3.214

2.4.1. Possible factors causing boiling-induced ageing

Boiling-induced ageing can be attributed to the combined influence of the following two factors:

i. There is a possibility of release of trapped air or oxygen from the scratches, cavities, and crevices on the copper surface during boiling [33].
ii. During the process of collecting the DI water from the source or while filling it in the rig's water tank, there is a possibility of absorption of CO_2 by water, which leads to the formation of carbonic acid which is a weak acid. Carbonic acid partially dissociates into hydrogen ions (H^+) and bicarbonate ions (HCO_3^-) ions or carbonate ions (CO_3^{2-}). This enables transfer of electrons and accelerates oxidation of copper [34,35].

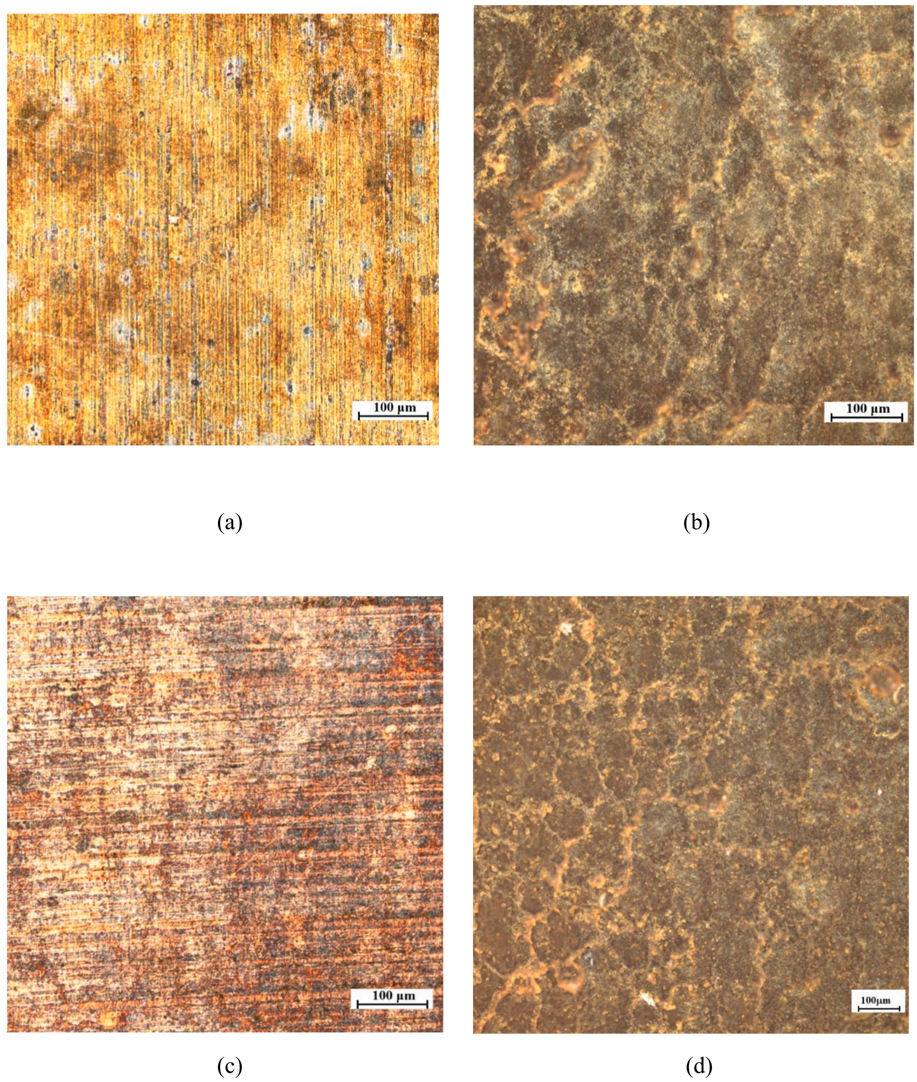


Fig. 5. Confocal microscope images of both channels. (a) 0.952 mm channel before ageing, (b) 0.952 mm channel after ageing, (c) 1.19 mm channel before ageing, (d) 1.19 mm channel after ageing.

3. Results and discussion

3.1. Single-phase friction factor validation

For friction factor validation of the 0.952 mm and 1.19 mm channels, single-phase experiments were conducted. Experimental friction factor was estimated for the Reynolds numbers in the range of 200 to 950. Theoretical friction factor was predicted employing the Shah [30] correlation and the values were compared. The results of the validation study are presented in Fig. 10. The Mean absolute error (MAE) between the experimental values and the predicted values of friction factor for the 0.952 mm channel before ageing and after ageing are 3.2% and 1.7%, respectively, which are within the uncertainty limit. Similarly, for the other channel also, the deviations are within the uncertainty limit.

3.2. Ageing study

Ageing runs were performed using the two channels. Boiling runs were repeated till the relative difference between HTCs of three consecutive runs was less than 10%. Mass fluxes of 500 kg/m^{2}s and 300 kg/m^{2}s were used for the 0.952 mm and 1.19 mm channels, respectively. The results are shown in Figs. 11 and 12. A maximum deterioration in two-phase HTC of 52% and 53% was observed for the 0.952 mm channel (at a heat flux of 1520 kW/m^2) and 1.19 mm channel (at a heat flux of 800 kW/m^2), respectively. The deterioration in two-phase HTC observed in the present study is in line with the findings by Ramakrishnan et al. [24], who reported a maximum reduction in two-phase HTC of 70% for a channel of hydraulic diameter 0.641 mm at a heat flux of 2450 kW/m^2 due to ageing. The plots showing variation of heat flux with wall superheat and variation of two-phase HTC with outlet quality for both channels are shown in Figs. E5 and E6 (in e-component). A plot showing variation of heat flux with the difference between average wall temperature and liquid inlet temperature for the 0.952 mm and 1.19 mm channels is shown in Figs. E7 and E8 (presented in e-component), respectively. The point of change of slope of the graph is used to distinguish the two-phase and single-phase and regions. It can be seen that the effect of ageing on wall temperature is prominent only in the two-phase region. Moreover, since the copper oxide layer thickness is only 7.06 µm, the maximum temperature drop across the layer at a heat flux of 4354 kW/m^2 is 0.92 °C assuming that the thermal conductivity of copper oxide is 32.9 W/mK [36]. This indicates that the observed large increase in wall temperature due to the formation of oxide layer is not caused by the thermal resistance offered by the layer. The reading of the thermocouple embedded in the wall accounts for the temperature drop resulting from (1) the conduction resistance across the copper wall, (2) the conduction resistance due to oxide layer on the channel surface, and

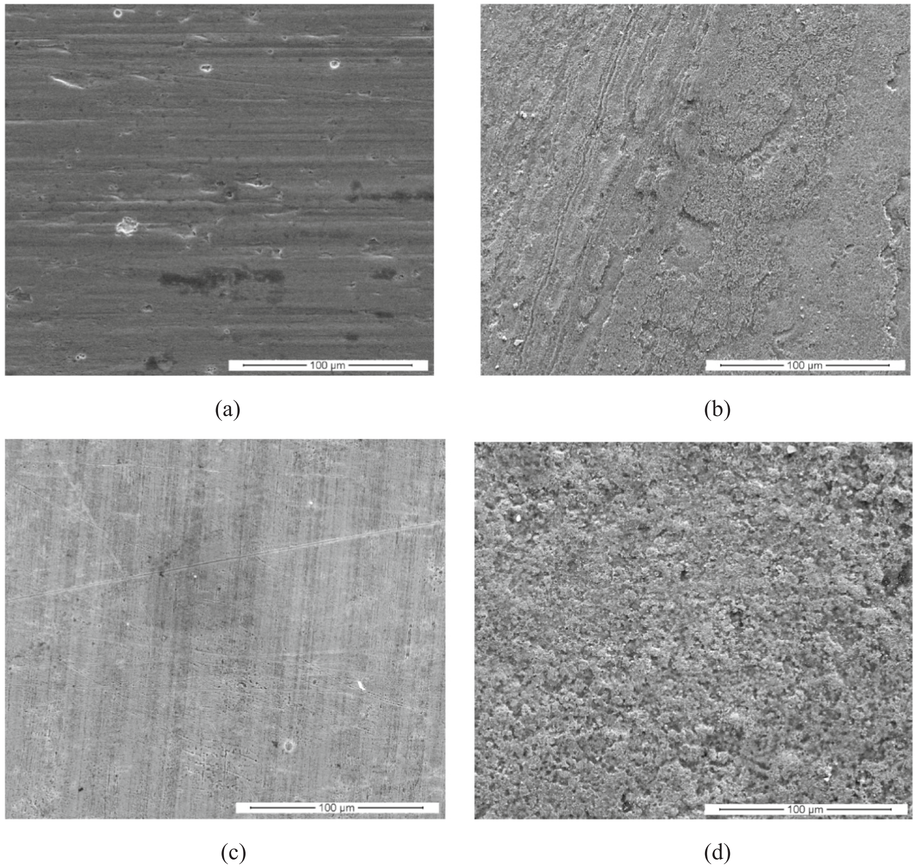


Fig. 6. SEM images of both channels. (a) 0.952 mm channel before ageing, (b) 0.952 mm channel after ageing, (c) 1.19 mm channel before ageing, (d) 1.19 mm channel after ageing.

(3) the convection resistance due to flow boiling. Two-phase heat transfer coefficient, determined using the wall temperature from Eq. (7) accounting for the temperature drop across the copper wall for the applied heat flux, is a result of the combined effect of the convection resistance due to flow boiling and the conduction resistance due to oxide layer on the channel surface.

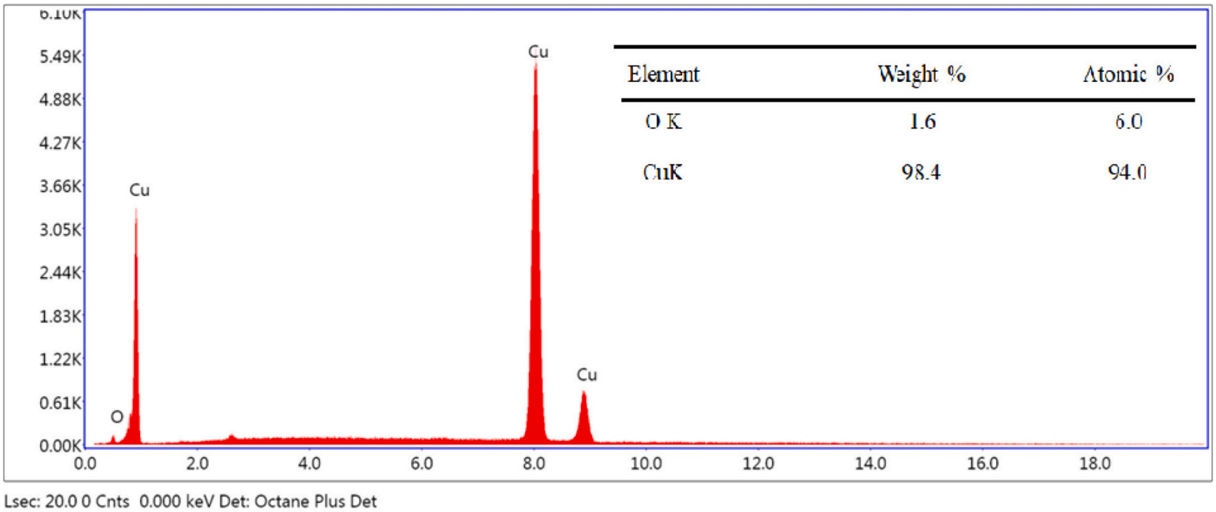
The apparent thermal conductivity of copper with oxide layer comes out to be 376.88 W/mK, so the conductivity changes from 385 to 376.88 W/mK. If the two-phase heat transfer coefficient is evaluated using the temperature of the oxide surface that comes in contact with the fluid, the heat transfer coefficient will be higher than that calculated using the channel wall temperature from Eq. (7). This is because the temperature of the oxide surface that comes in contact with the fluid will be lower than the channel wall temperature calculated from Eq. (7), or because the apparent thermal conductivity of copper with oxide is lower in Eq. (7). For a heat flux of 4354 kW/m^2, the variation in the evaluated two-phase HTC due to the thermal resistance of oxide layer is 4.9% and that for a heat flux of 413 kW/m^2 is 1%. It must be noted that the thermal resistance due to oxide layer has not been considered in determining two-phase HTC for the ageing runs as the measured oxide layer thickness of 7.06 μm is based on the measurement after the final ageing run. The oxide layer thickness is expected to change during the runs, with most of the oxide growth occurring during the first few initial runs, and the measurements have not been made after each run.

The reduction in HTC (increase in wall superheat) due to ageing can be explained by the increase in wettability. Nucleation occurs from pits or cavities on the heated surface and their dimensions determine the wall superheat needed for nucleation. Nucleation from a smaller pit or cavity requires a larger wall superheat. At the same time, if the surface wettability increases, it leads to flooding of larger cavities by liquid. The smaller cavities will not get flooded due to the resistance offered by surface tension. This prevents the larger cavities from trapping vapour and hence they can no longer act as active nucleation sites. The remaining smaller cavities which can act as active nucleation sites need a higher superheat for nucleation and hence cause the reduction in HTC.

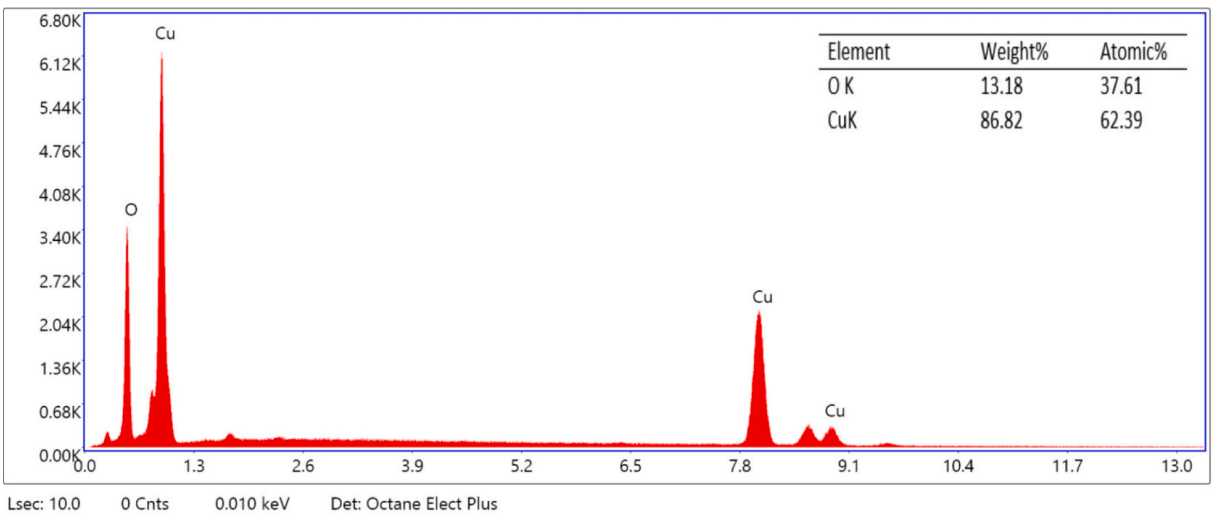
The maximum deteriorations in two-phase HTC for the channels of diameters 0.952 mm and 1.19 mm are 52% and 53%, respectively. The maximum reduction in two-phase HTC caused by ageing for the channel of hydraulic diameter 0.641 mm reported by Ramakrishnan et al. [24] is 70%.The higher reduction in two-phase HTC for the 0.641 mm channel is possibly due to the presence of stronger nucleate boiling dominant mechanism as compared to the other two channels of diameters 0.952 mm and 1.19 mm. With regard to the extent of possible influence of gases trapped in cavities or scratches of the substrate on boiling heat transfer coefficient, a discussion is presented in [24]. Large reductions in boiling heat transfer coefficient have been reported for pool boiling too. For nucleate pool boiling of water on a copper surface, a reduction of nearly 56% in heat transfer coefficient was observed in eight hours under boiling conditions (Jakob and Fritz, 1931, as cited in Collier and Thome [37], pp. 489–490).

The heat flux associated with the maximum HTC in the plot showing variation of HTC with heat flux is identified as the critical heat flux (CHF). This method is followed because a reduction in two-phase HTC with an increase in heat flux is caused by dryout or partial dryout in the case of saturated flow boiling. A similar method was followed by Kim and Mudawar [38] for identifying CHF data from the literature.

The change in CHF during the ageing experimental runs of the first channel is shown in Fig. 13. The results indicate that CHF is enhanced due to ageing. An average enhancement of 85% in CHF is observed considering CHF of the first run and the average CHF of runs 13, 14 and 15. The change in CHF during the ageing experimental runs of the second channel is shown in Fig. 14. The results indicate an enhancement in



(a)



(b)

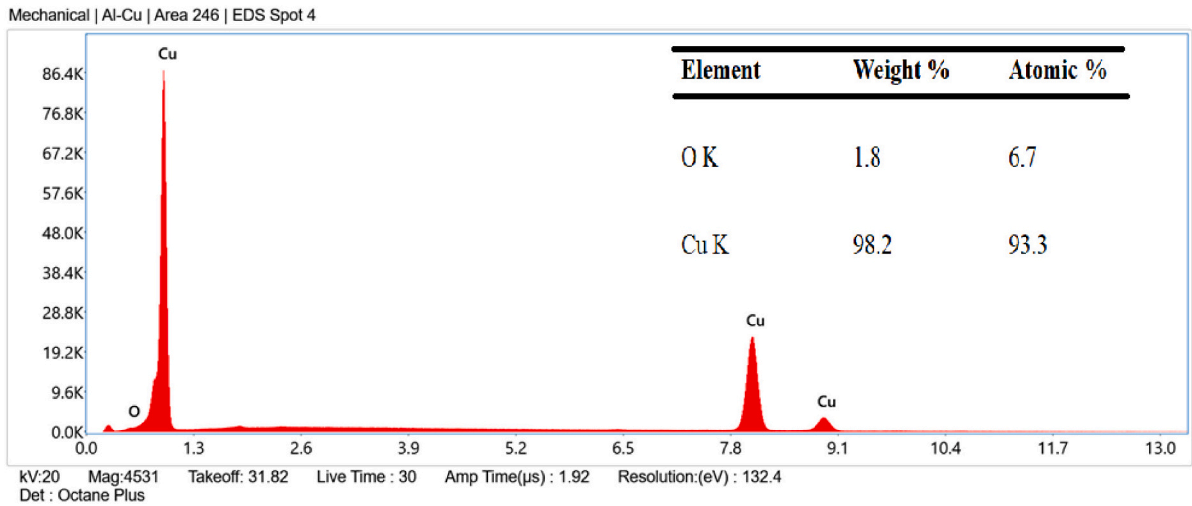
Fig. 7. EDS analysis of copper surface. (a) 0.952 mm channel before ageing (b) 0.952 mm channel after ageing, (c) 1.19 mm channel before ageing, (d) 1.19 mm channel after ageing.

CHF due to ageing similar to the first test section. An average enhancement of 78% in CHF is observed considering CHF of the first run and the average CHF of runs 3, 4 and 5. The enhancement of CHF observed due to ageing in the case of the 0.952 mm and 1.19 mm channels was not observed during the ageing experiments conducted by Ramakrishnan et al. [24] using the 0.641 mm channel. They observed no significant variation in CHF due to ageing for the 0.641 mm channel. The mean value of CHF of all the experimental runs was 1570 kW/m^2 and the standard deviation was 175 kW/m^2. The difference in CHF values among the runs was attributed to the following reasons:

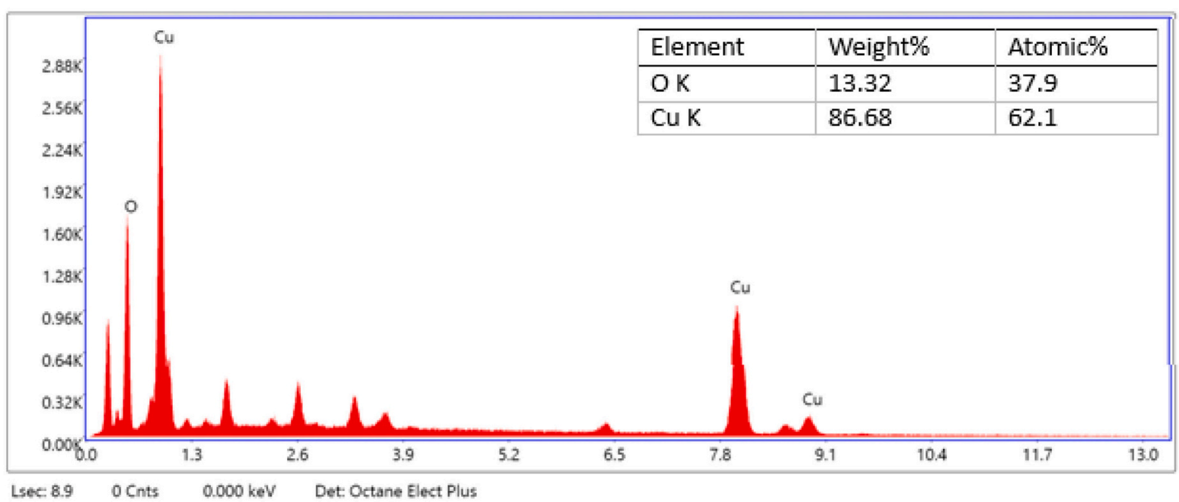
(i) Uncertainties related with point at which flow pattern transitions occur.
(ii) Uncertainties associated with identification of the point with maximum heat transfer coefficient.
(iii) Significant heat conduction along axial direction of channel caused by high thermal conductivity of copper material. This leads to a dampening of any local temperature overshoot and makes it less discernible.

Contact angle measurement results suggest that the wettability of all the three channels increased due to ageing. There are many investigations on flow boiling in microchannels which indicate that CHF will be enhanced due to an increase in wettability [1,4,39–42]. This is due to the fact that the increase in wettability helps to prevent or delay dryout of the thin liquid film in contact with the heated surface. The enhancement in CHF in the case of 0.952 mm and 1.19 mm channels are in line with the observations reported in the above-mentioned literature. At the same time, there are also some studies which point to the possibility that wettability may not influence CHF in all cases. A few possible explanations for the unaffectedness of CHF with increase in wettability observed for the 0.641 mm channel [24] are as follows:

(i) Mass fluxes employed for experiments with the 0.641 mm channel may have been sufficiently large to make the

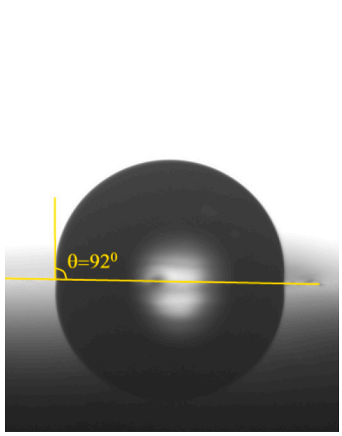


(c)

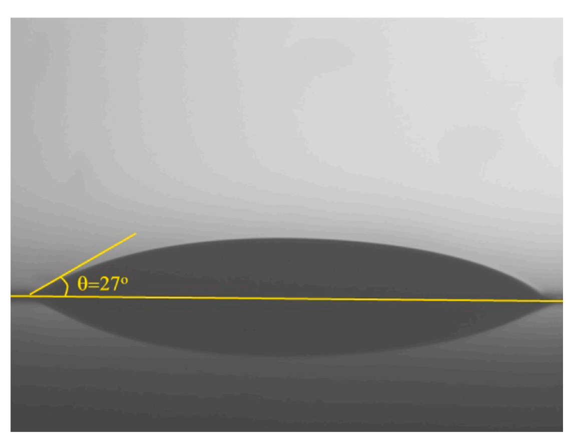


(d)

Fig. 7. (*continued*).



(a)



(b)

Fig. 8. Droplet placed on (a) Fresh channel and (b) Aged channel.

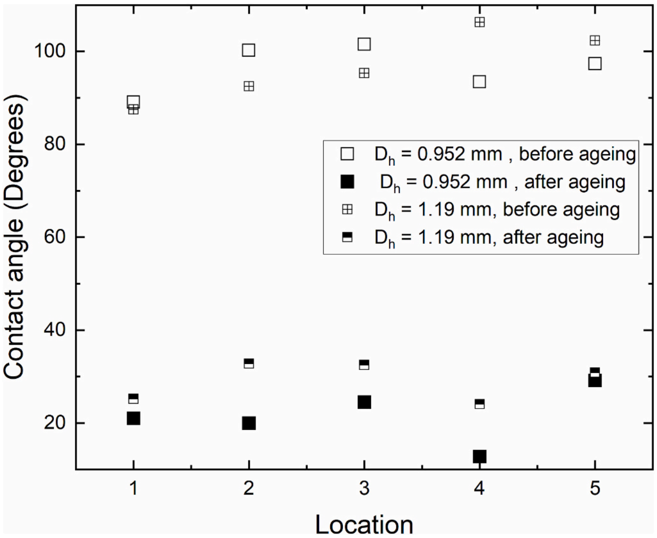


Fig. 9. Change in contact angle on the channel copper surface prior to and after ageing experiments.

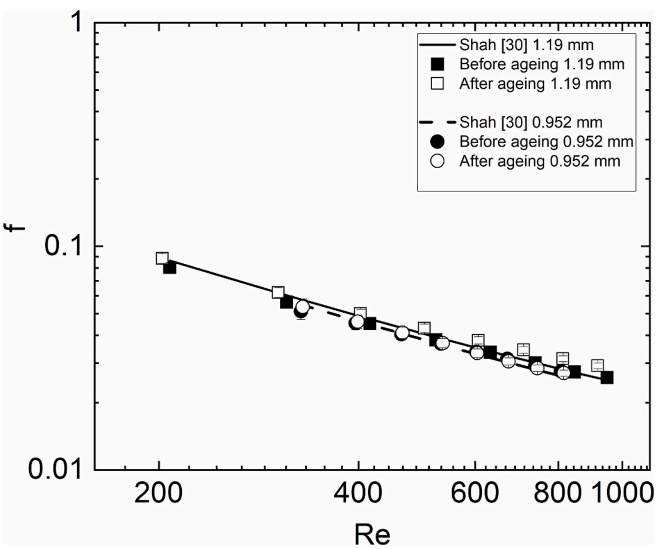


Fig. 10. Variation of single-phase friction factor with Reynolds number for the 0.952 mm and 1.19 mm channels.

contribution of liquid supply resulting from increased wettability insignificant (Park and Jeong [43]).

(ii) Flow patterns that occur within the range of experimental conditions studied can lead to a lower frequency of rewetting of the channel heated surface (Nascimento et al. [44]).

The combined effect of change in diameter of the channels along with surface characteristics would have influenced the CHF values.

The influence of ageing on CHF during flow boiling in the 0.952 mm and 1.19 mm channels was an enhancement in CHF unlike the 0.641 mm channel where CHF remained unaffected. To confirm whether the trend of an enhancement in CHF holds good for a different mass flux, ageing experiments were repeated on the 1.19 mm channel alone at a mass flux of 400 kg/m^{2}s. The results of the confirmatory ageing study are presented in e-component Figs. E9-E11. The change in CHF during ageing runs is shown in Fig. 14. The results indicate that the CHF is enhanced by up to 91% due to ageing. At the same time, a maximum decrease in two-phase HTC of 58% is observed between the first and last ageing runs. These results of the confirmatory ageing experiments are in line with the

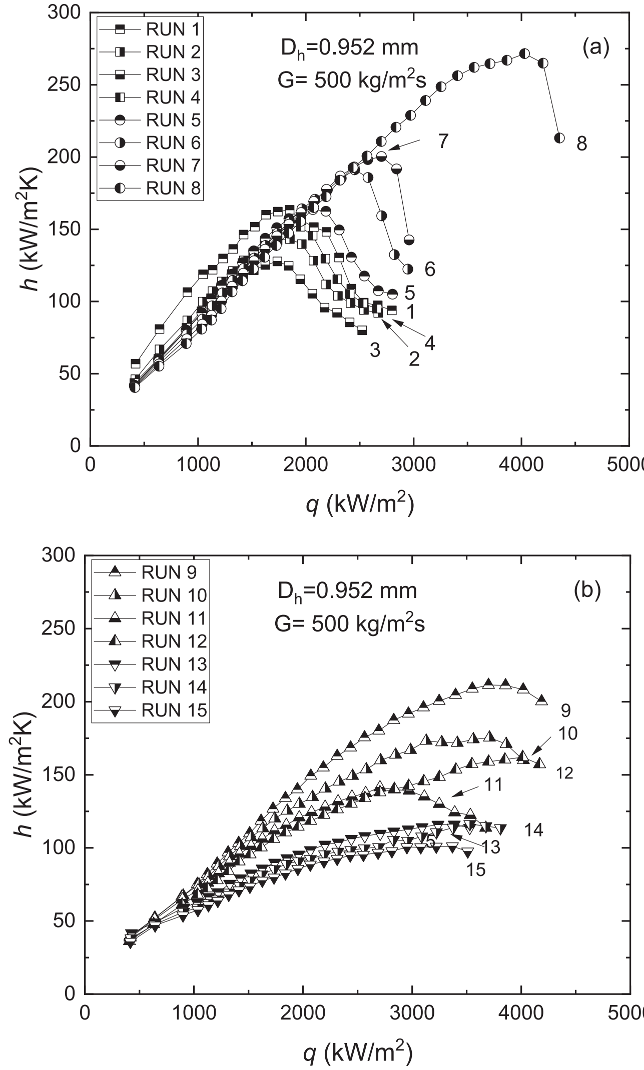


Fig. 11. Variation of two-phase HTC with heat flux for ageing runs of the first test section. (a) Runs 1–8 (b) Runs 9–15.

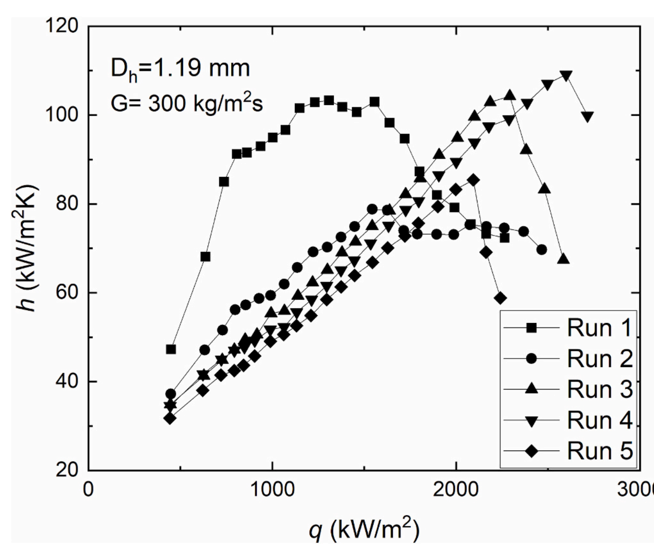


Fig. 12. Variation of two-phase HTC with heat flux for ageing runs of second test section.

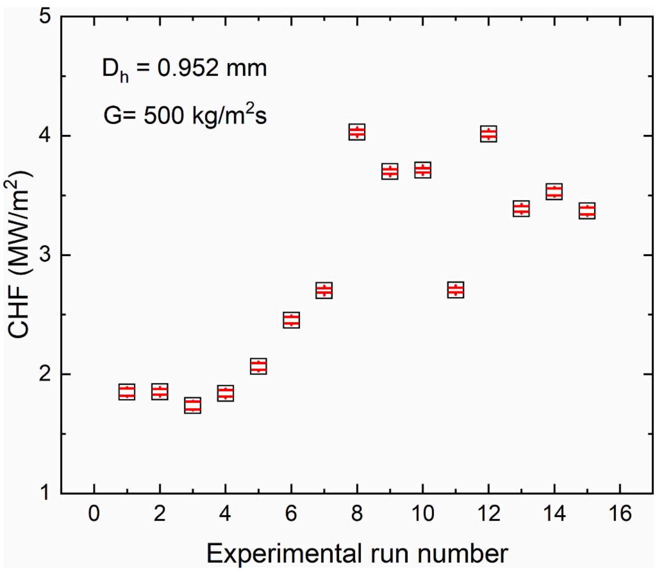


Fig. 13. Change in CHF during ageing experimental runs of the 0.952 mm channel.

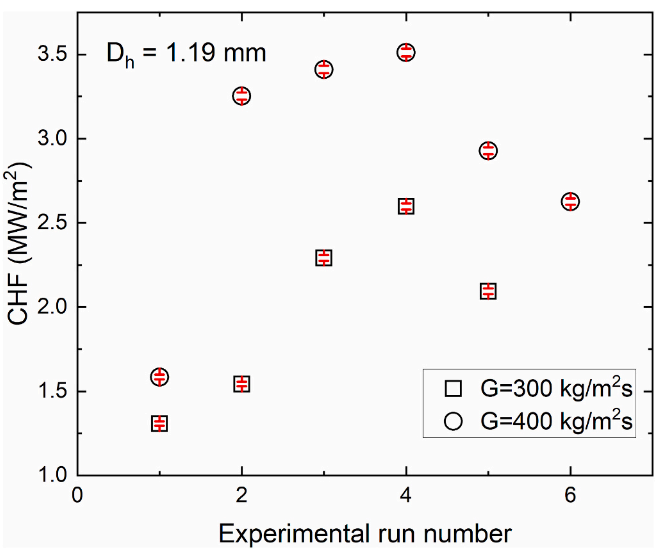


Fig. 14. Change in CHF during ageing experimental runs of the 1.19 mm channel.

results of ageing experiments carried out using the 0.952 mm channel and also the 1.19 mm channel at the mass flux of 300 kg/m^{2}s.

3.3. *Effect of mass flux on two-phase HTC and CHF*

Fully aged channels were used to carry out experiments for investigating the influence of mass flux on HTC and CHF. Mass fluxes of 300 and 400 kg/m^{2}s were employed for the 0.952 mm channel while mass fluxes of 200 and 400 kg/m^{2}s were employed for the 1.19 mm channel. The results for the 0.952 mm and 1.19 mm channels are shown in Fig. 15 and Fig. 16, respectively. Two-phase HTC was found to be repeatable within two experimental runs for the channels. The mean absolute deviation between two-phase HTC values of two runs for all the channels was within 4%. However, repeatability of CHF could be obtained only after four experimental runs for the 0.952 mm channel. The CHF increased with an increase in mass flux.

In the case of the 0.952 mm channel, the CHF values for the mass

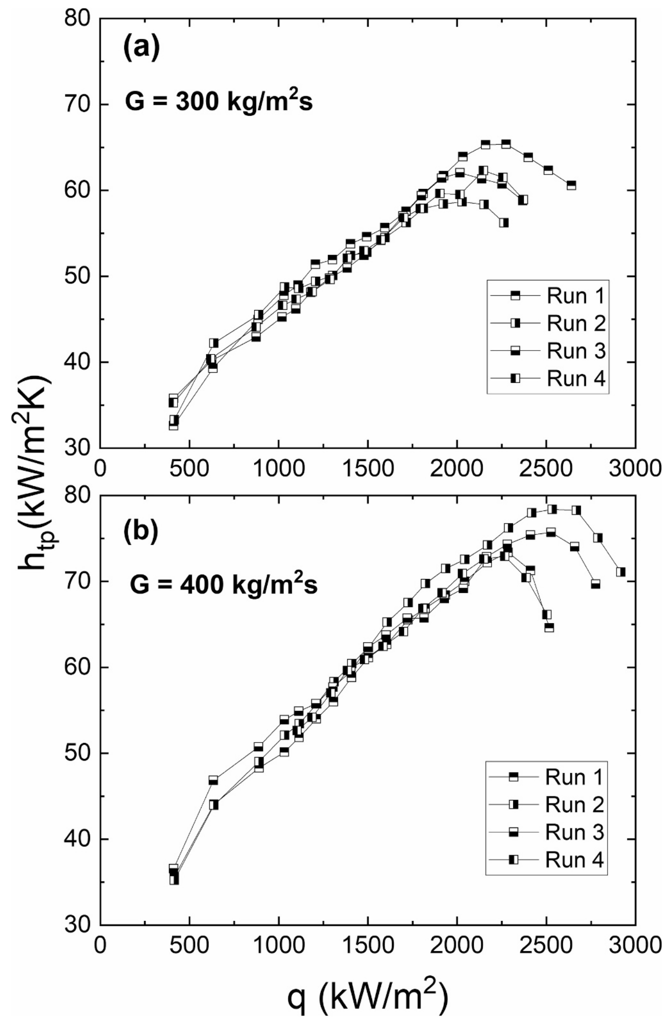


Fig. 15. Variation of two-phase HTC with heat flux for the aged 0.952 mm channel for different mass fluxes (a) 300 kg/m^{2}s (b) 400 kg/m^{2}s.

fluxes of 400 and 300 kg/m^{2}s are 2403 and 2117 kW/m^2, respectively. Average CHF of ageing runs 13, 14 and 15 conducted using the 0.952 mm channel at a mass flux of 500 kg/m^{2}s is 3429 kW/m^2.

For the 1.19 mm channel, average CHF values of 3018 and 1807 kW/m^2 were observed for the mass fluxes of 400 and 200 kg/m^{2}s, respectively. Mean CHF of ageing runs 3, 4 and 5 conducted using the 1.19 mm channel at a mass flux of 300 kg/m^{2}s is 2328 kW/m^2.

For the 0.641 mm channel, Ramakrishnan et al. [24] reported mean CHF values of 1570, 1926 and 2214 kW/m^2 for the mass fluxes of 700, 900 and 1100 kg/m^{2}s, respectively. Also, an average CHF of 1570 kW/m^2 was reported considering all ageing runs conducted at the mass flux of 500 kg/m^{2}s. The two-phase HTC in the case of the 0.641 mm channel was not affected significantly by mass flux and increased with heat flux till CHF. This indicated a nucleate boiling dominant behavior.

For the 0.952 mm channel the two-phase HTC is found to increase with both heat flux and mass flux. The mean deviation between two-phase HTC values at the mass fluxes of 400 kg/m^{2}s and 300 kg/m^{2}s is 11%. Similarly, an increase in two-phase HTC with both mass flux and heat flux is observed for the 1.19 mm channel. The mean deviation between two-phase HTC values at the mass fluxes of 400 kg/m^{2}s and 200 kg/m^{2}s is 20%. The dependence of HTC on mass flux in the case of 0.952 mm and 1.19 mm channels may be possibly due to the effect of channel size and flow patterns on the HTC. The plots showing variation of heat flux with wall superheat and variation of two-phase HTC with outlet quality for both channels are presented in Figs. E12-E15 (in e-

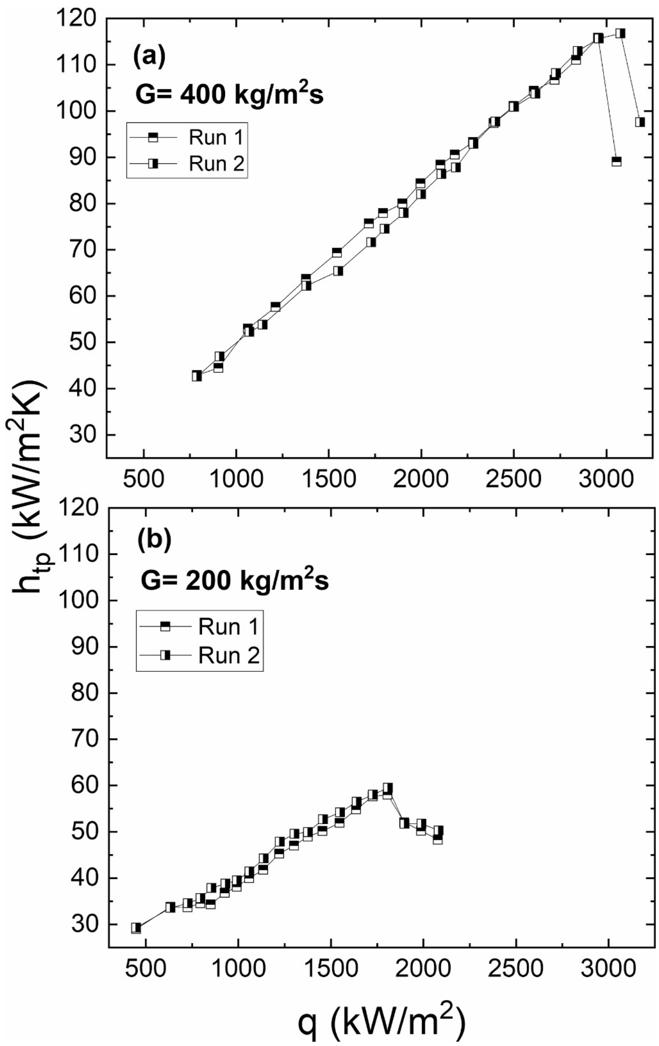


Fig. 16. Variation of two-phase HTC with heat flux for the aged 1.19 mm channel for different mass fluxes (a) 400 kg/m^{2}s (b) 200 kg/m^{2}s.

component). It should be noted that the gear pump characteristics caused the mass flux through the channel to decrease during the experiment. This decrease can be attributed to an increase in the two-phase pressure drop across the channel with an increase in the heat flux applied. In the course of the experimental runs, the variation of mass flux with heat flux is presented in Fig. E16 (in e-component).

3.4. Two-phase pressure drop

The two-phase pressure drops for the 0.952 mm channel and the 1.19 mm channel for different mass fluxes are shown in Fig. 17. The ageing experiments were performed at a mass flux of 500 kg/m^{2}s for the 0.952 mm channel and the results are shown in Fig. 17 (a) for Runs 1,2,14 and 15. The results indicate that the two-phase pressure drop for the ageing runs are nearly equal till their corresponding CHF. The pressure drops for Runs 1 and 2 increase after CHF possibly because of a change in flow pattern. Experiments at the mass fluxes of 300, 400 kg/m^{2}s reveal that an increase in mass flux results in an increase in two-phase pressure drop. This increase in two-phase pressure drop can be attributed to higher frictional pressure drop at higher mass fluxes. The ageing experiments were performed at a mass flux of 300 kg/m^{2}s for the 1.19 mm channel and the results are shown in Fig. 17 (b) for Runs 1,2,4 and 5. Similar to the 0.952 mm channel, two-phase pressure drop for the ageing runs are nearly equal till their corresponding CHF. The pressure

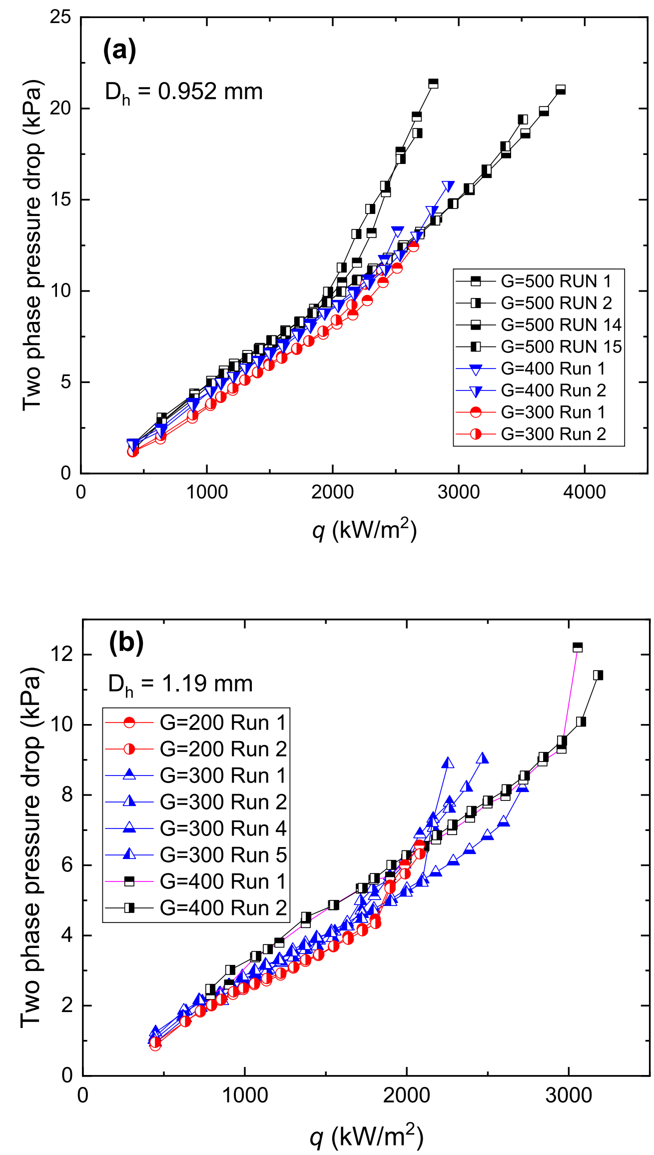


Fig. 17. Two-phase pressure drop versus heat flux for (a) 0.952 mm channel, (b) 1.19 mm channel.

drops for Runs 1,2 and 5 increase after CHF possibly because of a change in flow pattern. It can be seen that an increase in mass flux results in an increase in two-phase pressure drop for the 1.19 mm channel also.

3.5. Comparison of the aged channel HTC obtained from the freshly machined channel and the HCl cleaned channel

Fig. 18 shows a comparison of the two-phase HTCs in the fully aged 1.19 mm channel at a mass flux of 400 kg/m^{2}s obtained from two different initial surface conditions. They correspond to the freshly machined channel (Fig. 18(a)) and the 0.1 M HCl cleaned channel (Fig. 18(b)). The latter refers to the channel obtained after removing the oxide layer of the aged channel with 0.1 M HCl solution. It can be seen that the difference between two-phase HTCs of the two cases is negligible. This may be attributed to the fact that the average roughness values of the two aged channels are 1.323 μm and 1.132 μm, respectively, which are close to each other. It can be inferred from the HTC plots that the fully aged channel surface characteristics on the whole are nearly independent of the initial condition or the method to make the channel fresh.

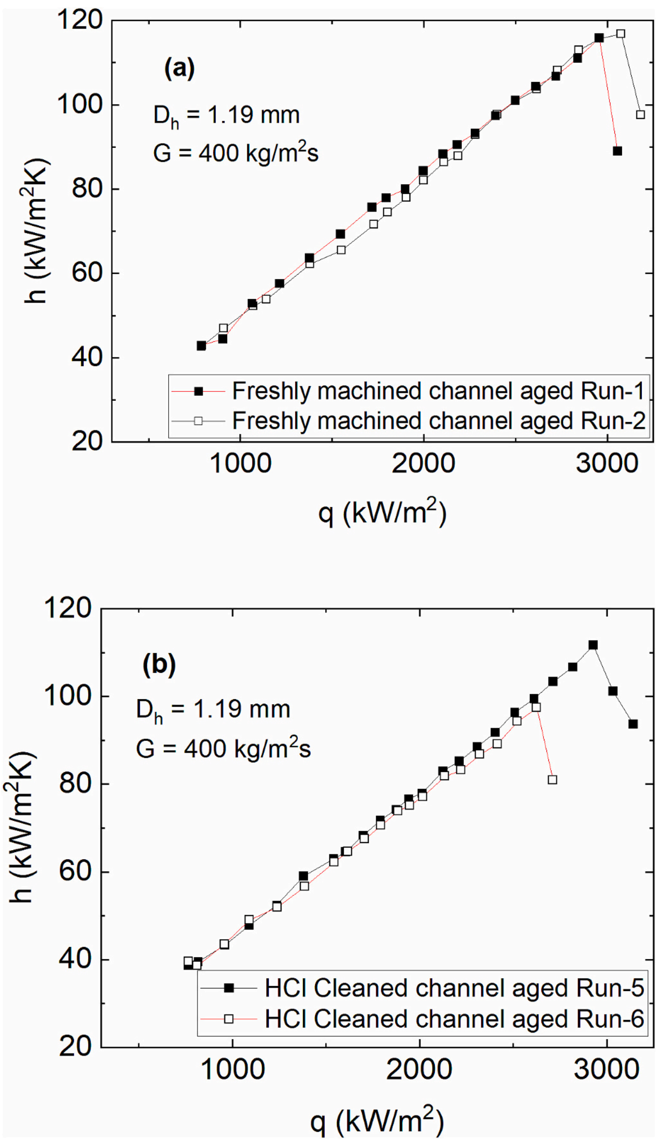


Fig. 18. Variation of HTC with heat flux for the fully aged 1.19 mm channel with different initial surface conditions, (a) freshly machined, (b) 0.1 M HCl cleaned.

4. Conclusions

1. A detailed experimental investigation of flow boiling of water in copper microchannels was conducted to investigate the effect of boiling induced ageing on the channel surface characteristics and the consequent heat transfer and CHF.
2. Number of experimental runs required to achieve the repeatability of the boiling curve (up to heat fluxes close to CHF) for a fresh copper channel were in the range of 5–15, the maximum for the 0.952 mm channel. And that for 0.1 M HCl cleaned channel was 6.
3. The maximum reductions in two-phase HTC for the 0.952 mm and 1.19 mm channels were 52% and 53%, respectively. The repeated boiling runs caused the surface roughness to increase from 0.259 μm to 0.656 μm and from 0.168 μm to 1.323 μm for the 0.952 mm and 1.19 mm channels respectively. SEM images and confocal microscope images revealed a distinct surface morphology change because of oxidation. EDS analysis revealed that the oxygen content (atomic percentage) increased on the 0.952 mm and 1.19 mm channels from 6% to 37.6% and from 6.7% to 37.9%, respectively. The average static contact angle reduced from 96^o to 22^o and from 97^o to 29^o for the 0.952 mm and 1.19 mm channels, respectively,
4. The deterioration in two-phase heat transfer coefficient is caused by an increase in wall superheat for nucleation due to the higher wettability of the channel surface resulting from oxidation.
5. The CHF was found to be enhanced due to ageing for the 0.952 mm (500 kg/m^{2}s) and 1.19 mm (300 kg/m^{2}s) channels by 85% and 78%, respectively. The enhancement can be attributed to the higher wettability due to ageing. Confirmatory experiments using 1.19 mm (400 kg/m^{2}s) channel also showed an enhancement in CHF by 91%.
6. However, in an earlier ageing study by the authors for the 0.641 mm channel [24], the CHF was not affected despite an increase in wettability. The possible reasons discussed were higher mass fluxes which render the influence of increased liquid delivery due to higher wettability insignificant and frequency of rewetting by liquid slugs not being sufficient due to the flow pattern.
7. The broad conclusion is that the CHF enhancement as a result of increased wettability due to ageing depends on the combined effect of channel mass flux and channel size. Higher mass fluxes could not be used for higher diameter channels due to the constraints of heater power (heat fluxes) and higher wall temperature associated with higher CHF for larger channels.
8. The increase in CHF with an increase in mass flux was corroborated by the results of the investigation on the influence of mass flux on CHF.
9. The influence of ageing on two-phase pressure drop was found to be insignificant when compared with the influence on two-phase heat transfer coefficient till CHF.

Part 2 of this study [45] makes an assessment of the existing correlations for two-phase heat transfer coefficient and CHF.

CRediT authorship contribution statement

Rishi L. Ramakrishnan: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sateesh Gedupudi:** Writing – review & editing, Visualization, Supervision, Project administration, Methodology, Formal analysis, Conceptualization. **Sarit K. Das:** Supervision, Resources, Project administration, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

Table A1
Correlations employed for single-phase friction factor validation.

Reference	Friction factor correlation	Condition
Shah and London [46]	$f_{fd} = \frac{24}{Re}\left(1 - 1.3553\alpha + 1.9467\alpha^2 - 1.70120\alpha^3 + 0.9564\alpha^4 - 0.2537\alpha^5\right)$	Laminar, fully developed flow.
Shah [30]	$f_{app} = \frac{3.44}{Re\sqrt{L^*}} + \frac{(fRe)_{fd} + \frac{K(\infty)}{4L^*} - \frac{3.44}{\sqrt{L^*}}}{Re\left(1 + \frac{C}{L^{*2}}\right)}$ $K(\infty) = 0.6611 + 1.1182\alpha + 2.1758\alpha^2 - 5.8322\alpha^3 + 4.4683\alpha^4 - 1.1553\alpha^5$ $L^* = \frac{L}{Red_h}; C = 0.0001966; Re = \frac{\rho V d_h}{\mu}$	Laminar, developing flow. f_{fd} calculated using [46].

Appendix B. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.icheatmasstransfer.2025.110139.

Data availability

Data will be made available on request.

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